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Prof ethics and social impact

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Project Proposal: PrivacyLens

I would like to create a Google Chrome extension that detects and explains how websites collect, track, and share users' personal information. We will call it PrivacyLens. While we do already have many extensions that do similar things, a lot of the browser tools only block trackers, PrivacyLens will take a more educational approach, it will translate complex technical behaviors like cookies, analytics scripts, and fingerprinting into plain-language descriptions that ordinary users can understand. The extension will also compare the site's privacy policy with the real data that it collects from users, helping users see when consent practices are misleading or incomplete.

I chose this project because all the discussions about ethical hacking in class made me realize how important data truly is, and how easily what we give away online can be used, manipulated, or sold without our awareness. The idea of ethical hacking shows that understanding systems not to exploit them, but to expose weaknesses responsibly is essential to keeping your digital information safe and free of exploitation from companies or websites.

I already know how to build browser extensions using HTML, CSS, and JavaScript, and I am familiar with Chrome's webRequest and cookies APIs for monitoring network activity. From the textbook (white hat), I understand the importance of responsible disclosure, which will guide how the extension analyzes sites without storing or transmitting sensitive data.

My project will focus on finding out how existing privacy tools work and understanding gaps in user comprehension. I will design a prototype that fills those gaps. After development, I will test PrivacyLens on different websites and make any adjustments if necessary.